

XIAOHE YIN

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EDUCATION

New York University

Aug. 2023

Master of Applied Urban Science and Informatics (GPA: 3.9/4)

New York, NY

- Coursework: Big Data Management, Principles of Urban Informatics, and Urban Spatial Analytics

Hong Kong Baptist University

May 2022

Bachelor of Science in Data Science (GPA: 3.68/4)

Hong Kong

- Coursework: Advanced Statistics, Data Mining, Data Visualization, Machine Learning, and Deep Learning

SKILLS

Analytical Skills: Statistical Modeling, Machine Learning, Data Visualization, Natural Language Processing (NLP), Large Language Models (LLMs), Full-stack Developments

Programming & Tools: ArcGIS, Python, R, JavaScript, HTML/CSS, Excel, PowerPoint, Power BI, Tableau

Languages: Chinese (Mandarin), Chinese (Cantonese), English

WORK EXPERIENCE

Research Assistant

Aug. 2025 – Jan. 2026

The University of Hong Kong, Department of Geography

Hong Kong

- Consolidated a transit dataset for 2000+ bus routes by integrating disparate open data from 3 major bus agencies.
- Automated **spatial database construction** with 22 independent variables based on Hong Kong network segments.
- Visualized aggregated weather and traffic data by network segments across time using **ArcGIS**.
- Managed end-to-end data pipeline and authored technical documents in terms of data definitions and processing logic.

Research Assistant

Dec. 2024 – Jun. 2025

The Hong Kong Polytechnic University, Department of Building and Real Estate

Hong Kong

- Developed an automated data pipeline using Python to crawl and clean dynamic hazard-related datasets.
- Constructed an **urban resilience ontology** to define complex components to support evidence-based disaster management.
- Leveraged **DeepSeek-V3** (LLM) with advanced prompt engineering to extract structured insights from news data.
- Implemented validation using semantic similarity metrics to ensure the accuracy and reliability of AI-extracted information.
- Visualized properties into **Neo4j knowledge graph** and **GIS-based maps** map to identify patterns across resilience cycle.

Data Science Intern

Oct. 2023 – Apr. 2024

United Nations, Business Transformation and Accountability Division

New York, NY

- Collaborated with cross-functional teams to design client-centric data product mock-ups and facilitated feedback sessions.
- Engineered data entity extraction process using **NLP** and **OCR** to digitize 200+ unsearchable PDF internal reports.
- Developed **Power BI** dashboard with geo-visuals and trend analysis across 47 civic categories to translate complex datasets.
- Spearheaded **full-stack web development** featuring **LLMs-driven Chatbot** and AI-summarization to ensure digitalization.
- Authored user documentation and technical manuals to ensure smooth handover, support maintenance, and guide end-users.

Graduate Research Assistant

Dec. 2022 - Apr. 2023

New York University Tandon School

New York, NY

- Implemented **ETL** pipeline to construct NYC building features datasets from 2500+ inspection reports.
- Developed an entity extraction model to identify facade components and defects, achieving 93% accuracy.
- Identified 4 NYC building groups and their defect patterns with **geo-spatial clustering analysis** (DBSCAN).

PROJECTS

NYC Electric Vehicles Adoption and Infrastructure Equity Consideration | *Python, PostgreSQL, Tableau*

- Aggregated sociodemographic and EV data from Open Data and organized state-level EV incentives.
- Constructed **Tableau dashboard** to visualize EV adoption trends and market share across boroughs.
- Conducted spatial autocorrelation using **Moran's I** to evaluate disparities in EV charging station accessibility.
- Identified socioeconomic structure on underserved communities using **spatial lag regression** and **k-means clustering**.

Sound Sources Identification in NYC Subway System | *Python, Javascript, GeoAnalytics*

- Designed movement detection using frame differencing to identify key timestamps across 14 subway Point-of-View videos.
- Implemented **object detection and segmentation** to analyze station-specific clips, identifying pedestrian density.
- Employed **time series analysis** to explore correlations between crowd density, train movement, and noise intensity.
- Delivered data insights covering noise levels, crowd density, and station 3D map using an **interactive map**.